

Teacher Guide — senior / module-09-subtraction-nikhilam

IKS Curriculum

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Overview

Subtraction — Nikhilam (All from 9)

Band: Senior (Grades 9–12) · **Sessions:** 10×45 min · **Sanskrit:** Nikhilam Navataścaramaṁ Daśataḥ

Overview

Nikhilam Navataścaramaṁ Daśataḥ — “all from 9, the last from 10” — is Tirthaji’s second sūtra. It produces fast complements to powers of 10, removing the need for borrowing in many subtractions. Conceptually identical to two’s-complement binary arithmetic used in modern computers.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential learning
- §11.8 — Holistic, multidisciplinary (math + computing)

Key terms (this module)

- *Nikhilam Navataścaramaṁ Daśataḥ* — “all from 9, the last from 10”
- *pratiyogi* (modern coinage) — the number that adds to a base

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 ($\times 11$ Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper
- (See `lesson-plan.md` for per-day material lists)

10-day map

- **Day 1** — The Sūtra — All from 9, Last from 10: Compute the complement of any 2-digit number to 100.
- **Day 2** — Complement to 1000 and Beyond: Compute complements to 1000 and 10000 mentally.
- **Day 3** — Subtract from a Power of 10: Compute 1000–567 and similar instantly.
- **Day 4** — General Subtraction via Complements: Use add-the-complement trick for arbitrary subtractions.
- **Day 5** — Why It Works — The Algebra: Derive the place-value identity that makes the rule work.
- **Day 6** — Bar Model Proof: Draw a visual proof using bar models and number lines.
- **Day 7** — Two's Complement in Computers: Show how the same idea underlies binary computer subtraction.
- **Day 8** — Speed Day: Complete a 30-problem mental subtraction sprint.
- **Day 9** — Mixed Practice: Solve a mixed problem set comparing complement vs traditional borrow.
- **Day 10** — Final Assessment: Take the summative quiz.

Files in this pack

- `lesson-plan.md` — detailed 10-day arc
- `sources.md` — sources cited in this module
- `teacher-notes.md` — common misconceptions, differentiation
- `parent-guide.md` — what parents should know
- `days/day-NN-*.md` — 10 per-day lesson plans
- `quizzes/formative.md`, `quizzes/summative.md`
- `activities/activity-01-*.md`
- `assessments/rubric.md`, `assessments/project-brief.md`
- `student-workbook.md`
- `writing-prompts.md` (or `oral-recall.md` for Primary)
- `homework.md`
- `slides/deck.md`

Voice

Analytical, comparative, primary-source aware; 20–30 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — Subtraction — Nikhilam (All from 9) (Senior)

10-day arc, 45 min per session.

Day 1 — The Sūtra — All from 9, Last from 10

Objective: Compute the complement of any 2-digit number to 100.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Complement to 1000 and Beyond

Objective: Compute complements to 1000 and 10000 mentally.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Subtract from a Power of 10

Objective: Compute 1000–567 and similar instantly.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — General Subtraction via Complements

Objective: Use add-the-complement trick for arbitrary subtractions.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Why It Works — The Algebra

Objective: Derive the place-value identity that makes the rule work.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Bar Model Proof

Objective: Draw a visual proof using bar models and number lines.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Two's Complement in Computers

Objective: Show how the same idea underlies binary computer subtraction.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Speed Day

Objective: Complete a 30-problem mental subtraction sprint.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Mixed Practice

Objective: Solve a mixed problem set comparing complement vs traditional borrow.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Assessment

Objective: Take the summative quiz.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — Subtraction — Nikhilam (All from 9) (Senior)

Common student misconceptions

- *“This is religious, not scientific.”* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *“The Sanskrit terms are just jargon.”* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *“If it’s traditional, it must work.”* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask “how would I test this?”.

Differentiation strategies

- **Extension for advanced students:** offer one optional “deeper read” per day — usually a primary source or modern paper.
- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition’s framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the “Vedic” status of Tirthaji’s mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn’t?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — Subtraction — Nikhilam (All from 9) (Senior)

Primary and secondary sources

- Bharati Krishna Tirthaji, *Vedic Mathematics* (1965), sūtra 2
- S. G. Dani, “Myths and reality: On Vedic mathematics,” *Frontline* (1993)
- NCERT Mathematics Class V, ch. 1 (The Fish Tale — subtraction)
- D. E. Knuth, *The Art of Computer Programming*, Vol. 2 (on complements in computing)

A note on citation discipline

Per the curriculum’s style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Senior-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at [curriculum/_shared/glossary.md](#) contains the IAST + meaning for every Sanskrit term used.
- Other modules in this curriculum that connect: see the module README's "Prerequisites" section.

Day 1 — The Sūtra — All from 9, Last from 10

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute the complement of any 2-digit number to 100.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: Nikhilam Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Primary source (today's reading)

From *Bharati Krishna Tirthaji* — see [sources.md](#).

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What's one thing you remember about yesterday's lesson?*”
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **Pose the question.** “*What's 100 – 47?*” Most students compute 53 by borrowing. Ask: “*What's 1000 – 437?*” Trickier — the borrowing chains. Today's method removes the chain entirely.
2. **State the sūtra.** *Nikhilam Navataścaramaṁ Daśataḥ* — “all from 9, the last from 10.” Write the Sanskrit and IAST on the board.
3. **Apply to 100 – 47.** Subtract each digit of 47 from 9, except the last which is from 10:
 - 4 from 9 = 5

- 7 from 10 = 3
 - Answer: **53**. No borrowing.
4. **Pattern recognition.** For $1000 - 437$: $9-4=5$, $9-3=6$, $10-7=3 \rightarrow$ **563**. Verify: $437 + 563 = 1000$. ✓
5. **Source note.** Tirthaji's *Vedic Mathematics* (1965), sūtra #2. Honest history: this is one of the more solidly-rooted methods in older Indian and Arabic arithmetic traditions (decimal complement was widely used pre-modern). Independent practice: 10 problems on the worksheet.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces The Sūtra — All from 9, Last from 10.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 2 — Complement to 1000 and Beyond

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute complements to 1000 and 10000 mentally.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (modern coinage) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define complement.** The complement of N to base B is $B - N$. Today we find complements to 1000 mentally.
2. **Worked examples.**
 - $738 \rightarrow 9-7=2, 9-3=6, 10-8=2 \rightarrow \mathbf{262}$. Check: $738 + 262 = 1000$. ✓
 - $405 \rightarrow 9-4=5, 9-0=9, 10-5=5 \rightarrow \mathbf{595}$. Check: $405 + 595 = 1000$. ✓
 - $999 \rightarrow 0, 0, 1 \rightarrow \mathbf{001}$. Edge case: trailing 9s carry differently — “*all from 9 except the LAST non-zero digit, which is from 10.*”
3. **The “trailing zero” caveat.** For 730 (ends in 0): treat the last non-zero digit as the one from 10. $730 \rightarrow 9-7=2, 10-3=7, 0 \rightarrow \mathbf{270}$. Or equivalently: complement 73 to 100 (= 27), append the trailing 0 $\rightarrow 270$.
4. **Pattern algebra.** For $N < 10^k$, the complement is $(10^k - 1) - N + 1 = (10^k - N)$. The “all from 9 last from 10” procedure computes this digit-by-digit.
5. **Practice.** 15-problem mental sprint, complements to 1000 only. 5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Complement to 1000 and Beyond.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Subtract from a Power of 10

Module: Subtraction — *Nikhilam* (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute $1000-567$ and similar instantly.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: *Nikhilam Navataścaramaṁ Daśataḥ*; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The clean application.** Whenever we’re subtracting from 10^k (10, 100, 1000, 10000, ...), *Nikhilam* gives the answer directly. No borrowing, no carries.
2. **Speed examples.**
 - $10 - 4 = 6$ (4 from 10)
 - $100 - 37 = 63$ (3 from 9, 7 from 10)
 - $1000 - 268 = 732$
 - $10000 - 1547 = 8453$
3. **Practical use case.** Computing change at a shop. “*You paid ₹1000 for an item costing ₹427. How much change?*” *Nikhilam*: $9-4=5$, $9-2=7$, $10-7=3 \rightarrow ₹573$. Mental math, no paper.

4. **Verify with traditional subtraction.** Have students cross-check 2 problems by the borrowing method. Same answer. The Nikhilam is just a smarter procedure for the same arithmetic.
5. **Class discussion.** “*When IS Nikhilam genuinely faster than borrowing? When does borrowing keep up?*” (Borrowing keeps up for small numbers like 10–4. Nikhilam dominates for larger powers.)

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Subtract from a Power of 10.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — General Subtraction via Complements

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: use add-the-complement trick for arbitrary subtractions.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pratiyogi (modern coinage)* (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Beyond powers of 10.** Most real subtractions aren’t from a clean 10^k . So how does Nikhilam help with, say, $6347 - 2589$?
2. **The “add-the-complement” trick.** Rewrite: $6347 - 2589 = 6347 + (10000 - 2589) - 10000 = 6347 + (\text{complement of } 2589) - 10000$.
 - Complement of 2589 to 10000: $9-2=7$, $9-5=4$, $9-8=1$, $10-9=1 \rightarrow 7411$.
 - $6347 + 7411 = 13758$.
 - $13758 - 10000 = 3758$.
 - Check: $6347 - 2589 = 3758$. ✓
3. **When is this faster?** When the borrowing in standard subtraction would chain through many digits, the complement method skips it entirely. For simple subtractions, standard is fine.
4. **Worksheet.** 8 problems, mixed: some clean Nikhilam, some general using add-the-complement.
5. **Reflection.** “*Which method-type did you find genuinely faster? Which felt the same?*” Honest answers — Nikhilam is a tool, not a universal speed-up.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces General Subtraction via Complements.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 5 — Why It Works — The Algebra

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive the place-value identity that makes the rule work.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: *Nikhilam Navataścaramaṁ Daśataḥ*; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the identity.** For N with k digits, $10^k - N$ can be computed as $((10^k - 1) - N) + 1$.
2. **Why “all from 9 last from 10” works.**
 - $(10^k - 1)$ is a number with k nines: 9, 99, 999, 9999, ...
 - $(10^k - 1) - N$: each digit of N is subtracted from 9. The result has no borrows because each digit of $(10^k - 1)$ is 9.
 - Adding 1 to that bumps the last digit by 1, which is equivalent to “the last digit from 10” rather than “from 9.”

3. **Worked algebraic example.** $N = 437$, $k = 3$.
 - $999 - 437 = 562$ (digit-wise: 9-4, 9-3, 9-7)
 - $562 + 1 = 563 = 1000 - 437$. ✓
4. **The general formula.** $10^k - N = (10^k - 1) - N + 1$ — a foundational identity in modular arithmetic and binary computer arithmetic.
5. **Senior bridge.** This is the same identity that underlies *two's complement* in binary computer arithmetic — tomorrow's topic.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Bar Model Proof

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: draw a visual proof using bar models and number lines.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (modern coinage) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Draw a bar of length 100.** Cut a section of length 37 from the right end. Label the remaining bar.
2. **Decompose using 99.** Show that the bar of 100 = bar of 99 + bar of 1. The bar of 99 is easy to subtract from (every digit is 9).
3. **Visual mapping to digits.**
 - Top row: 99 partitioned into a section of 37 and a section of $(99-37) = 62$.
 - Bottom row: add the unit 1, splitting it into the rightmost cell.
 - Result: $(99-37)+1 = 100-37 = 63$.
4. **Generalise visually.** Show the same bar-decomposition for $1000-437$. The leading 9s allow digit-wise subtraction without borrow; the +1 at the end bumps the last digit.
5. **Independent task.** Each student draws their own bar-model proof for $1000 - 248 = 752$ and submits as proof of understanding.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Bar Model Proof.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 7 — Two's Complement in Computers

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: show how the same idea underlies binary computer subtraction.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: Nikhilam Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Bridge to computing.** Modern computers don’t have a subtraction circuit — they have an ADDER circuit. They subtract by adding the *two’s complement* of the subtrahend. This is the same Nikhilam idea, in binary.
2. **Binary Nikhilam.** For an 8-bit number:
 - One’s complement: flip every bit (= “all from 1,” the binary analogue of “all from 9”).
 - Two’s complement: one’s complement + 1.
3. **Worked example.** Compute $9 - 5$ in 8-bit binary:
 - $5 = 0000101$
 - Flip: 11111010 (one’s complement)

- Add 1: 11111011 (two's complement)
 - Add to 9 = 00001001: $00001001 + 11111011 = 100000100$, drop the overflow bit
→ $00000100 = 4$. ✓
4. **Source.** D.E. Knuth, *The Art of Computer Programming*, vol. 2 — full treatment.
 5. **Critical-thinking close.** *“The Nikhilam sūtra (or its decimal-complement principle) and binary two’s complement both date their origins to vastly different traditions and eras. What does this tell us about (a) the universality of certain algorithmic ideas and (b) the limits of cultural-priority claims?”*

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Two’s Complement in Computers.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 8 — Speed Day

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: complete a 30-problem mental subtraction sprint.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (modern coinage) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **30-problem mental sprint.** Mix:
 - Pure Nikhilam (subtract from 10, 100, 1000, 10000) — 15 problems
 - Add-the-complement (general subtractions) — 10 problems
 - Edge cases (trailing zeros, single-digit) — 5 problems
2. **5-minute timer.** Silence. Pen on paper.
3. **Pair-mark.** Swap and check.
4. **Distribution analysis.** “*Where did errors cluster?*” Most likely: edge cases (trailing zeros), large complements (10000 case).
5. **Personal best.** Record score; set target for next sprint.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Speed Day.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.

- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 9 — Mixed Practice

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: solve a mixed problem set comparing complement vs traditional borrow.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: Nikhilaṁ Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Mixed problem set.** 20 subtractions, classes mixed:
 - Simple borrows (use borrow or Nikhilam, your choice)
 - Multi-borrow (Nikhilam likely faster)
 - Subtraction from powers of 10 (Nikhilam dominates)
 - Add-the-complement (general)
2. **Strategy choice.** Each student picks the method per problem and writes the chosen method beside the answer.
3. **Cross-check.** Pairs swap; compare method choices. “*Did you both pick the same method? When did you disagree?*”
4. **Discussion.** “*Is Nikhilam ALWAYS better? When is plain borrowing fine?*” (Single-digit, small subtractions.)
5. **Method-comparison artefact.** Each student starts a “personal mental-math toolkit” page summarising when to use each method.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Mixed Practice.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Final Assessment

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: take the summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (*modern coinage*) (IAST: *pratiyogi* (*modern coinage*); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Reflection essay (10 min).** *“In 200 words, explain Nikhilam to a younger student. Use one worked example and one limit-case.”*
2. **Summative quiz (25 min).** Computation, algebraic proof, two’s-complement connection. See `quizzes/summative.md`.
3. **Computer-subtraction explainer project share (10 min).** Each student presents their poster (see `assessments/project-brief.md`).
4. **Module close.** *“What’s one number-sense habit from this module you’ll keep using?”* Each student shares one sentence.
5. **Linkage.** Tomorrow Module 11 builds the multiplication side of Nikhilam — same sūtra, different application.

Activity (15 min)

A scaled version of *Complement Flashcards* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Final Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Formative Quiz — Subtraction — Nikhilam (All from 9) (Senior)

When: mid-module (after Day 5). **Duration:** 25 minutes. **Format:** mix of short answer, multiple choice, and reflection.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Subtraction — Nikhilaṃ (All from 9) in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Subtract from a Power of 10’?
(a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Subtraction — Nikhilaṃ (All from 9) is purely a religious concept and has no scientific framing.*
5. Connect Subtraction — Nikhilaṃ (All from 9) to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘Nikhilaṃ Navataścaramaṃ Daśataḥ’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about Subtraction — Nikhilaṃ (All from 9) and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it’s a check-in to see what’s landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Subtraction — Nikhilaṃ (All from 9) (Senior)

When: end of module (after Day 10). **Duration:** 45 minutes. **Total points:** 30.

Questions

Part A — Vocabulary (~5 points) A1. Define *Nikhilaṃ Navataścaramaṃ Daśataḥ* in your own words. A2. Define *pratiyogi* (modern coinage) in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Subtraction — Nikhilaṃ (All from 9) in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Subtraction — Nikhilaṃ (All from 9) with a concept from another module (your choice). What’s similar, what’s different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Subtraction — Nikhilaṃ (All from 9) from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What's one question you still have?

Part E — Primary source (~5 points; Senior only) E1. Quote or paraphrase one passage from a primary source cited in this module. Explain it in your own words.

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 18/30 indicates the student has met the module's learning objectives.

Homework

Homework — Subtraction — Nikhilam (All from 9) (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sūtra — All from 9, Last from 10** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Complement to 1000 and Beyond** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Subtract from a Power of 10** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — General Subtraction via Complements** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Why It Works — The Algebra** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Bar Model Proof** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Two's Complement in Computers** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Speed Day** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Mixed Practice** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.

- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — Subtraction — Nikhilam (All from 9) (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Subtraction — Nikhilam (All from 9) to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Subtraction — Nikhilam (All from 9) with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Subtraction — Nikhilam (All from 9)? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 1 — Complement Flashcards

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of Subtraction — Nikhilam (All from 9) through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

40 flashcards (target – number) pre-made

What students do

In pairs, one student holds up “1000 – 738” (etc.), the other shouts the answer within 3 seconds using the Nikhilam method.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Computer Subtraction Explainer

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior

What you will do

Make a one-page explainer poster: “How does my calculator subtract?” Show two’s complement subtraction in binary using the same principle as Nikhilam.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Subtraction — Nikhilam (All from 9) (Senior)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

Accuracy of concepts — uses Sanskrit terms correctly, gets the central idea right

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Source discipline — cites sources, distinguishes traditional framing from empirical claim

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Application — connects the module to other modules and to real-world examples

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Presentation — clear, organised, evidence-supported

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Reflection — honest, thoughtful, identifies what was learned and what remains uncertain

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.